

Scalability & Future Plan

Date	2 October 2026
Team ID	SWTID-2026-1380
Project Name	PocketSmart AI: Your Smart Budget & Recommendation Assistant
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Current recommendation data may depend on configured AI/service integrations.	Live product/service information may not always be available.	High
2	Local application/database setup is suitable for a prototype-scale deployment.	Larger user loads would require stronger shared infrastructure.	Medium
3	Recommendation quality depends on input quality and AI response quality.	Incomplete context can reduce recommendation relevance.	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Prototype/local deployment	Use scalable application hosting, load balancing and horizontal scaling for larger user bases.
2	Data Storage	Local database for application data/history	Move to managed relational storage and add backup, indexing and retention policies.
3	Performance	FastAPI + AI request/response workflow	Add caching, asynchronous processing, request limits and performance monitoring.
4	Security	Session-based authentication and environment-based secrets	Strengthen authentication, secret management, access controls, HTTPS and security monitoring.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Strengthen live product/service integrations and recommendation evaluation.	Next development cycle	More current and measurable recommendations.
Phase 2	Richer personalization and user preference history.	Medium term	More relevant recommendations for returning users.
Phase 3	Additional planning and recommendation categories.	Medium to long term	Broader usefulness beyond the current three scenarios.

Phase 4	Broader deployment options and expanded multimodal experiences.	Long term	Improved scalability, accessibility and AI-assisted planning.
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